# Mengxuan Shen

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## Education

**The University of Manchester**
*MSc Robotics* *2026–2027*

**Xi'an Jiaotong-Liverpool University**
*BEng Digital Media Technology,* ***First-Class Honours*** *Aug. 2022–Jun. 2026*

- Relevant modules: Artificial Intelligence, Image Processing, Embedded Computer Systems, Microprocessor Systems, Computer Graphics, Signals and Systems, Electronic Circuits, Mobile Computing, Human-Centric Computing.
- Final-year thesis: *Multi-Head Learning for Computation Offloading in Mobile Edge Computing* (see Research).

## Internship Experience

**Vision System Assistant, Hebei Beifang Welding Tools Co., Ltd.**
*Jul. 2025–Sep. 2025*

- Addressed a quality-inspection bottleneck where manual checking of robotic welding seams took 3–5 minutes per electrode holder; proposed an image-based pre-screening tool to flag obvious defects before human verification.
- Collected 450 weld images with the workshop's industrial camera and wrote Python/OpenCV scripts to extract seam regions using edge detection and morphological operations; labelled crack, porosity and incomplete-fusion patterns as ground truth.
- With a senior engineer, implemented a defect-detection pipeline combining Canny edge detection (cracks) and contour analysis (porosity), tuning thresholds with feedback from two QC inspectors across 120 test cases.

**Defect Analysis Intern, Hebei Jincheng Yuan Electric Power Tools Co., Ltd.**
*Jul. 2024–Oct. 2024*

- Investigated a 30% false-alarm rate in a newly deployed automated inspection system for insulated tool handles by analysing 200 misclassified images; used histogram analysis to quantify brightness differences between true defects and false positives.
- Built a Python/OpenCV preprocessing pipeline comparing Gaussian, median and bilateral filtering; adaptive histogram equalisation with morphological closing reduced lighting-induced false alarms by 45% in controlled tests on 150 samples.

## Research Experience

**Multi-Head Learning for Edge Computing Offloading**
*Final-Year Thesis, Xi'an Jiaotong-Liverpool University* *2025–2026*

- Extended a shared-backbone multi-task network (MTFNN-CO baseline) with five specialised prediction heads to decide which mobile tasks to offload to an edge server (Python, TensorFlow/Keras).
- Designed an "enhanced gating" head selector using only deployment-available signals; it outperformed voting and input-only gating at every scale tested (e.g. 82.4% vs 79.1% decision accuracy at four users).

**Vision-Guided Robotic Control for Hand Rehabilitation**
*Research Assistant, Robotics Laboratory* *Oct. 2024–Feb. 2025*

- Developed a real-time visual feedback system for a rehabilitation robot that previously followed fixed trajectories, detecting hand keypoints with OpenCV and MediaPipe from an overhead camera.
- Designed a position-based visual servoing controller in Python, mapping joint coordinates from image space to the robot frame through calibration matrices to adjust trajectories to the patient's actual position.
- Integrated visual feedback into the control loop so the robot adapted when subjects shifted posture, reducing the need for manual therapist intervention.

**Deep Learning for Medical Image Enhancement**
*Research Assistant, Medical Imaging Group* *Feb. 2024–Jun. 2024*

- Built a U-Net enhancement model for degraded chest X-rays (poor contrast, motion blur, under-exposure), with a weighted loss emphasising anatomical boundaries and suspected abnormal regions, plus elastic and intensity augmentation.

**Edge Machine Learning for IoT Navigation System**
*Research Assistant, Embedded Systems Lab* *Jan. 2024–May 2024*

- Developed an on-device decision-tree model for a smart umbrella to recommend routes from weather patterns without cloud dependency, using GPS and environmental sensor data, deployable on Arduino hardware.

**Retrieval-Augmented Generation for Research Question Answering**
*Research Assistant under Prof. Philipp Koehn (Johns Hopkins University)* *May 2023–Sep. 2023*

- Built a hybrid retrieval pipeline (BM25 filtering, then Sentence-BERT embeddings with a FAISS index for semantic re-ranking) to cut query response time while keeping relevant papers in the top results.

## Project Experience

**Intelligent Umbrella for Rainy-Day Navigation**
*Team Leader, Jiangsu Provincial College Robot Creativity Competition —* ***Second Prize*** *Mar. 2025–Jun. 2025*

- Led a team of four to prototype a smart umbrella with LED lighting, NFC, GPS tracking and 4G connectivity; wrote Arduino C/C++ for NFC data exchange, GPS acquisition and 4G communication.
- Designed structural parts in CAD and 3D-printed iterative prototypes; managed task allocation, troubleshooting and the competition demonstration.

**AI Personal Assistant Mobile Application (AIVA)**

*UI/UX Designer* *Mar. 2025–May 2025*

- Designed the full interface for an AI assistant app (scheduling, diet recommendations, decision support) in Figma, from wireframes to high-fidelity interactive prototypes.

**Other Projects**

*Source code and demos: shine574269.github.io* *2024–2026*

- DailySense Android app (UI design and calendar module); TED Talks TF-IDF search and recommendation system; Unity 2D shooter (C#); 8051 embedded pulse counter (C); OpenGL graphics (C++); MATLAB image processing and eigenface recognition.
- Mathematical Contest in Modeling: led a three-member team through a 96-hour optimisation problem using Python, Java and MATLAB.

## Awards

- Second Prize, Jiangsu Provincial College Intelligent Robot Creativity Competition (Jun. 2025)
- Honorable Mention, Mathematical Contest in Modeling (MCM) (Feb. 2024)
- First Place, ELC Sustainable Development Video Creation Competition (Apr. 2023)
- Provincial Outstanding Class, Xi'an Jiaotong-Liverpool University (Jun. 2023)

## Skills

- **Programming:** Python, Java (proficient); C/C++, Embedded C, C#
- **AI & Computer Vision:** OpenCV, MediaPipe, PyTorch, TensorFlow; CNNs, U-Net, Transformers; visual servoing
- **Embedded & Hardware:** Arduino, 8051 (Keil), GPS/4G/NFC modules, CAD and 3D printing, MATLAB
- **Tools & Design:** Git, MySQL, Unity, Android Studio, Figma, Photoshop, Premiere

## Leadership & Activities

- Director of Administration & Finance, Student Union, XJTLU (Feb. 2023–Jun. 2024); Head of Publicity, International Innovation Society (Feb.–Nov. 2024); Class Monitor (2022–2023)
- Translator Volunteer, International Wushu Invitational Tournament (2024); All-Subject Tutor (2022–2025)